

SYDNEY GRAMMAR SCHOOL



2016 Half-Yearly Examination

FORM V

MATHEMATICS 2 UNIT

Tuesday 17th May 2016

General Instructions

- Writing time — 1 hour 30 minutes
- Write using black pen.
- Board-approved calculators and templates may be used.

Total — 80 Marks

- All questions may be attempted.

Section I – 8 Marks

- Questions 1–8 are of equal value.
- Record your answers to the multiple choice on the sheet provided.

Section II – 72 Marks

- Questions 9–14 are of equal value.
- All necessary working should be shown.
- Start each question in a new booklet.

Collection

- Write your name, class and Master on each answer booklet and on your multiple choice answer sheet.
- Hand in the booklets in a single well-ordered pile.
- Hand in a booklet for each question in Section II, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Write your name, class and Master on this question paper and hand it in with your answers.
- Place everything inside the answer booklet for Question Nine.

5A: DNW

5B: PKH

5C: LRP

5D: FMW

5E: WJM

5F: GMC

5G: NL

5H: SO

5P: TCW

5Q: SDP

5R: RCF

Checklist

- SGS booklets — 6 per boy
- Multiple choice answer sheet
- Candidature — 178 boys

Examiner

NL

SECTION I - Multiple Choice

Answers for this section should be recorded on the separate answer sheet handed out with this examination paper.

QUESTION ONE

Which of the following bearings is due west?

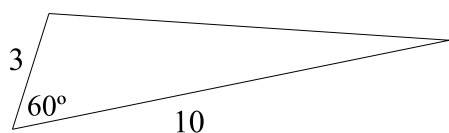
- (A) 000° T
- (B) 090° T
- (C) 180° T
- (D) 270° T

QUESTION TWO

What is the correct expansion of $(x - 1)(x + 4)$?

- (A) $x^2 - 3x + 4$
- (B) $x^2 + 4x - 4$
- (C) $x^2 + 3x - 4$
- (D) $x^2 + 4x - 3$

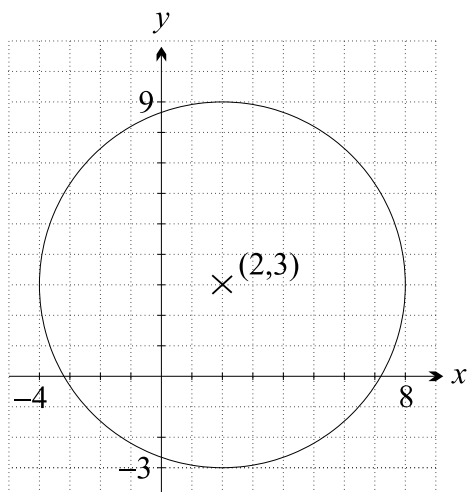
QUESTION THREE



Which expression is the exact area of the triangle above?

- (A) $5\sqrt{2}$
- (B) $\frac{20\sqrt{2}}{3}$
- (C) $\frac{15\sqrt{3}}{2}$
- (D) $10\sqrt{2}$

QUESTION FOUR



The equation of the circle drawn above is:

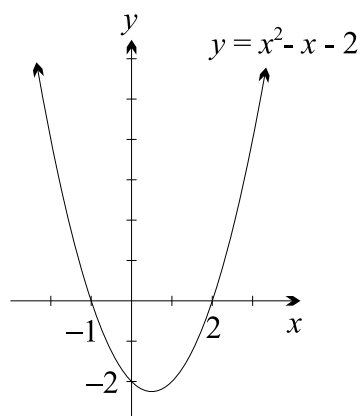
- (A) $(x + 2)^2 + (y + 3)^2 = 6$
- (B) $(x - 3)^2 + (y - 2)^2 = 6$
- (C) $(x + 3)^2 + (y + 2)^2 = 36$
- (D) $(x - 2)^2 + (y - 3)^2 = 36$

QUESTION FIVE

Which expression is equivalent to $x^2 + 4x + 9$?

- (A) $(x + 2)^2 + 13$
- (B) $(x - 2)^2 + 5$
- (C) $(x + 2)^2 + 9$
- (D) $(x + 2)^2 + 5$

QUESTION SIX



Using the graph above, the solution to the inequation $x^2 - x - 2 > 0$ is:

- (A) $-1 \leq x \leq 2$
- (B) $x < -1$ or $x > 2$
- (C) $-1 > x > 2$
- (D) $x < -1$ or $x < 2$

QUESTION SEVEN

Which of these functions is even?

- (A) $f(x) = x^2 - 8x + 15$
- (B) $f(x) = (x + 2)^2 - 4$
- (C) $f(x) = \sqrt{9 - x^2}$
- (D) $f(x) = x^3$

QUESTION EIGHT

Which of the following identities is not true?

- (A) $\sec^2 \theta + 1 = \cot^2 \theta$
- (B) $\sin^2 \theta + \cos^2 \theta = 1$
- (C) $1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$
- (D) $\tan^2 \theta + 1 = \sec^2 \theta$

_____ End of Section I _____

SECTION II - Written Response

Answers for this section should be recorded in the booklets provided.

Show all necessary working.

Start a new booklet for each question.

QUESTION NINE (12 marks) Use a separate writing booklet.	Marks
(a) If $f(x) = 4x - 3$, find $f(2)$.	1
(b) Evaluate $ -5 - -7 $.	1
(c) Write $\frac{2}{b} + \frac{1}{3}$ as a single fraction.	1
(d) Express $\frac{3}{2 - \sqrt{2}}$ as a simplified fraction with a rational denominator.	2
(e) What is the exact value of $\sqrt{27} + \sqrt{12}$?	1
(f) (i) Factorise $x^2 - x - 6$.	1
(ii) Hence solve $x^2 - x - 6 = 0$.	1
(g) Consider the points $A(6, 9)$ and $B(2, 1)$.	
(i) Find the coordinates of the midpoint of the interval AB .	1
(ii) Find the gradient of AB .	1
(iii) Find the equation of the line which passes through points A and B .	2

QUESTION TEN (12 marks) Use a separate writing booklet.

Marks

- (a) Show that the lines $y = 3x - 4$ and $x + 3y = 12$ are perpendicular. **2**
- (b) Solve $2x - 4y = -12$ and $3x + 2y - 2 = 0$ simultaneously. **2**
- (c) Consider the parabola with equation $y = x^2 - 6x + 7$.
- (i) Show that the parabola has x -intercepts $x = 3 + \sqrt{2}$ and $x = 3 - \sqrt{2}$. **1**
- (ii) Write down the y -intercept. **1**
- (iii) Write down the equation of the axis of symmetry. **1**
- (iv) Find the coordinates of the vertex. **1**
- (v) Sketch the graph of $y = x^2 - 6x + 7$, clearly marking all of the above features. **2**
- (d) Given that $\cos \theta = \frac{3}{7}$ and $\tan \theta > 0$, find $\sin \theta$. Leave your answer in exact form. **2**

QUESTION ELEVEN (12 marks) Use a separate writing booklet.

Marks

(a) (i) Solve $|x - 1| = 5$.

2

(ii) Solve $|3x + 2| > 1$.

2

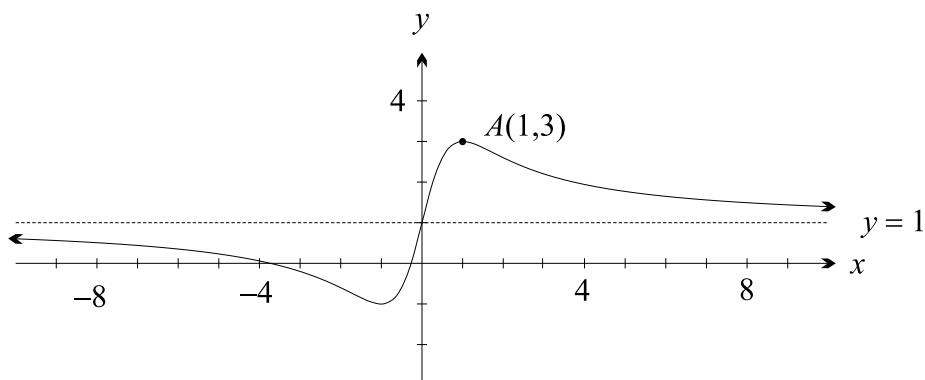
(b) Solve $\cos x = 0.6$, for $0^\circ \leq x \leq 360^\circ$. Give your answer to the nearest degree.

2

(c) Find the perpendicular distance from the point $P(2, 3)$ to the line $x + 2y + 3 = 0$.

2

(d)



The graph of $y = g(x)$ is shown above. It has horizontal asymptote $y = 1$. State the coordinates of the point A and the equation of the asymptote after each of the following transformations.

(i) $y = g(x) + 1$

2

(ii) $y = -g(x)$

2

QUESTION TWELVE (12 marks) Use a separate writing booklet.**Marks**

- (a) Sketch each function below on separate axes, including labelled asymptotes and intercepts with axes if they exist.

(i) $y = \sqrt{x + 4}$

1

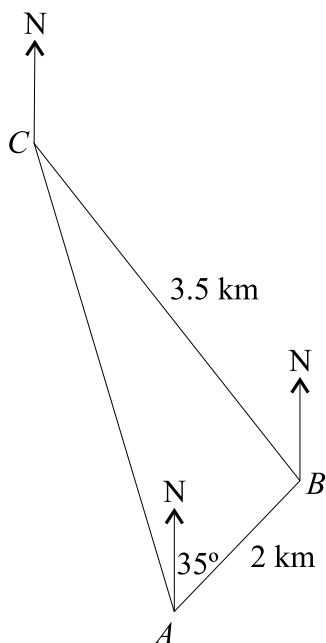
(ii) $y = |x + 5|$

2

(iii) $y = \frac{1}{x - 1} + 3$

3

(b)



The diagram above shows the path of a hiker. The hiker starts from point A and walks on a bearing of 035° T for 2 km to point B . She leaves point B on a bearing of 320° T and walks for 3.5 km until she reaches point C .

- (i) Show that $\angle ABC$ is 105° . Justify your answer with geometrical reasoning.
- (ii) Find the distance AC . Give your answer correct to the nearest metre.
- (iii) Find the bearing of C from A . Give your answer correct to the nearest degree.

2**2****2**

QUESTION THIRTEEN (12 marks) Use a separate writing booklet.

Marks

(a) Consider the function $f(x) = 2^x + 1$.

(i) Sketch the graph of $y = f(x)$.

2

(ii) State the range of the function.

1

(b) Find the angle of inclination of the line $y = -4x + 2$. Give your answer correct to the nearest degree.

2

(c) Solve $\cos 2\alpha = \frac{\sqrt{3}}{2}$, for $0^\circ \leq \alpha \leq 360^\circ$.

3

(d) Consider the points $A(3, 9)$ and $B(1, 1)$ and the circle with diameter AB .

(i) Show that the equation of the circle is $(x - 2)^2 + (y - 5)^2 = 17$.

2

(ii) The line $y = x + 6$ cuts the circle at point A and again at a second point C . Find the coordinates of the point C .

2

QUESTION FOURTEEN (12 marks) Use a separate writing booklet.**Marks**

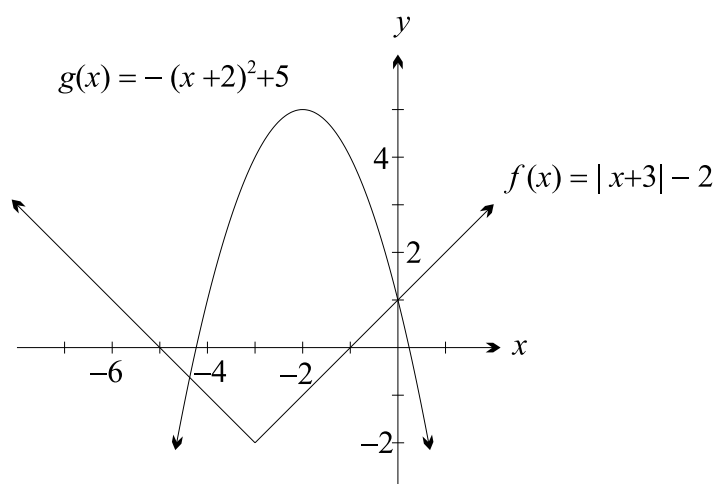
- (a) (i) Express
- $A^3 + B^3$
- as a product of two factors.

1

- (ii) Hence prove the identity
- $\frac{\sin^3 x + \cos^3 x}{\sin x + \cos x} = 1 - \sin x \cos x$
- .

2

(b)



Consider the functions $f(x) = |x + 3| - 2$ and $g(x) = -(x + 2)^2 + 5$ in the diagram above.

- (i) Show that
- $y = f(x)$
- and
- $y = g(x)$
- intersect at
- $(0, 1)$
- .

1

- (ii) Find the exact values of the coordinates of the second point of intersection of
- $y = f(x)$
- and
- $y = g(x)$
- .

3

- (iii) Hence solve
- $|x + 3| - 2 \leq -(x + 2)^2 + 5$
- .

1

- (c) Two points
- A
- and
- B
- lie on a line
- l
- . The midpoint of
- AB
- is
- $M(6, 7)$
- . The line
- $2x + 5y - 15 = 0$
- passes through
- A
- and the line
- $3x - 2y + 5 = 0$
- passes through
- B
- .

Let A have coordinates $A(h, k)$.

- (i) Show that the coordinates of
- B
- are
- $B(12 - h, 14 - k)$
- .

1

- (ii) Find the equation of
- l
- .

3

_____ End of Section II _____

END OF EXAMINATION

NAME:

CLASS: MASTER:

SYDNEY GRAMMAR SCHOOL



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- Record your multiple choice answers by filling in the circle corresponding to your choice for each question.
- Fill in the circle completely.
- Each question has only one correct answer.

Question One

A ☐ B ☐ C ☐ D ☐

Question Two

A ☐ B ☐ C ☐ D ☐

Question Three

A ☐ B ☐ C ☐ D ☐

Question Four

A ☐ B ☐ C ☐ D ☐

Question Five

A ☐ B ☐ C ☐ D ☐

Question Six

A ☐ B ☐ C ☐ D ☐

Question Seven

A ☐ B ☐ C ☐ D ☐

Question Eight

A ☐ B ☐ C ☐ D ☐

2016 Form V Mathematics Solutions.

- 1) D
- 2) C
- 3) C
- 4) D
- 5) D
- 6) B
- 7) C
- 8) A

Q9)

$$a) f(2) = 4 \times 2 - 3 \\ = 5$$

$$b) | -5 | - | -7 | \\ = -2$$

$$c) \frac{2}{b} + \frac{1}{3} \\ = \frac{6}{3b} + \frac{b}{3b} \\ = \frac{6+b}{3b}$$

$$d) \frac{3}{2-\sqrt{2}} \\ = \frac{3(2+\sqrt{2})}{(2-\sqrt{2})(2+\sqrt{2})} \quad \checkmark \\ = \frac{6+3\sqrt{2}}{2} \quad \checkmark \quad (\text{or factorised})$$

$$e) 3\sqrt{3} + 2\sqrt{3} = 5\sqrt{3}$$

$$f) i) x^2 - x - 6 = (x+2)(x-3)$$

$$ii) x = -2 \quad \text{or} \quad x = 3$$

$$g) i) M = \left(\frac{6+2}{2}, \frac{9+1}{2} \right) \\ = \left(\frac{8}{2}, \frac{10}{2} \right) \\ = (4, 5) \quad \checkmark$$

$$ii) m = \frac{9-1}{6-2} \\ m = 2 \quad \checkmark$$

$$iii) y - 1 = 2(x - 2) \quad \checkmark \\ y = 2x - 4 + 1 \\ y = 2x - 3 \quad \checkmark$$

Q10.

a) $y = 3x - 4$
 $m_1 = 3$

$$x + 3y = 12$$

$$3y = -x + 12$$

$$y = -\frac{1}{3}x + 4$$

either $\checkmark \rightarrow m_2 = -\frac{1}{3}$

$$\therefore 3 \times -\frac{1}{3} = -1 \quad \checkmark$$

or $m_1 \times m_2 = -1$ (if explicitly stated m_1 and m_2 above)

b) ① $2x - 4y = -12$

② $3x + 2y = 2$

② $6x + 4y = 4$

$$\begin{array}{r} \textcircled{2} + 1 \\ + \left(\begin{array}{r} 6x + 4y = 4 \\ 2x - 4y = -12 \\ \hline 8x = -8 \\ x = -1 \quad \checkmark \end{array} \right) \end{array}$$

($\checkmark$ for correctly forming an equation with one variable;
 $3(2y - 6) + 2y - 2 = 0$)

Sub: $-2 - 4y = -12$
 $y = \frac{5}{2} \quad \checkmark$

c) $y = x^2 - 6x + 7$

i) $a=1$ $b=-6$ $c=7$

$$x = \frac{6 \pm \sqrt{(-6)^2 - 4 \times 1 \times 7}}{2 \times 1}$$

$$x = \frac{6 \pm \sqrt{8}}{2}$$

$$x = 3 \pm \sqrt{2}$$

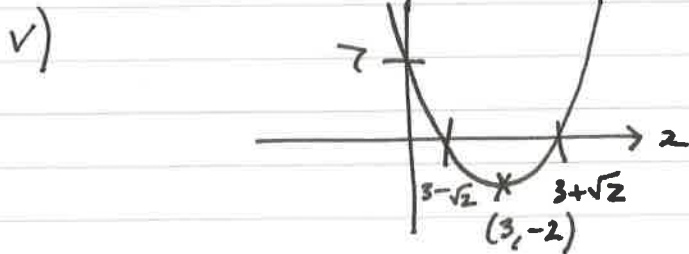
✓ $\left(\text{or } (x-3)^2 = 2 \quad \checkmark \right)$
 $x = 3 \pm \sqrt{2}$

ii) 7 or $(0, 7)$

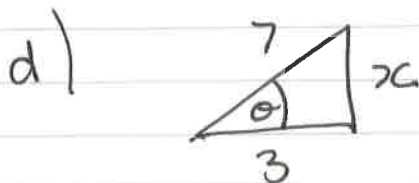
iii) $x = -\frac{b}{2a}$
 $= \frac{6}{2}$
 $= 3 \quad \checkmark$

iv) $y = 3^2 - 6 \times 3 + 7$
 $= -2$

$\therefore (3, -2) \quad \checkmark$



- 1 for missing -intercepts
- poor shape
- axis labels
- vertex label



$x = 2\sqrt{10} \quad \checkmark$ (or 1st quadrant ✓)

$\sin \theta = \frac{2\sqrt{10}}{7} \quad \checkmark$

(Accept $\frac{\sqrt{40}}{7}$)

Q11.

a) i) $x-1=5$

$x=6$ ✓

or $x-1=-5$

$x=-4$ ✓

(or other valid method 1 mark for working)

ii) $3x+2>1$

$3x>-1$

$x>-\frac{1}{3}$

or

$3x+2<-1$

$3x<-3$

$x<-1$

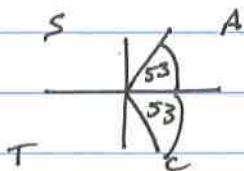
$\therefore x<-1$ ✓

or $x>-\frac{1}{3}$ ✓

b) $\cos x = 0.6$

$x \approx 53.13^\circ$ ✓

$x = 53^\circ$ (to nearest degree)



and $x \approx 306.86^\circ$

$x = 307^\circ$ ✓

(to nearest degree).

c) $P(2,3)$ $x+2y+3=0$

$x_1=2$ $y_1=3$

$a=1$ $b=2$ $c=3$

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

$$= \frac{|1(2) + 2(3) + 3|}{\sqrt{1^2 + 2^2}}$$
 ✓

$$= \frac{|11|}{\sqrt{5}}$$
 ✓

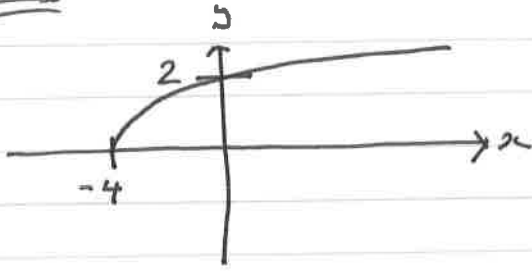
$$= \frac{11\sqrt{5}}{5}$$

d) i) $(1, 4)$ ✓
 $y = 2$ ✓

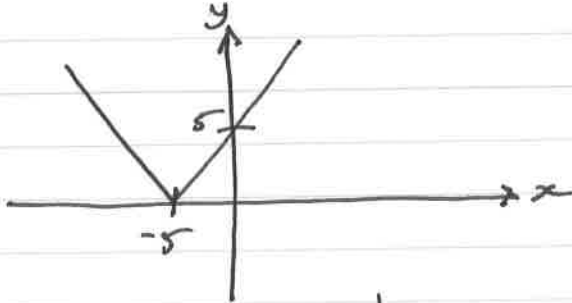
ii) $(1, -3)$ ✓
 $y = -1$ ✓

Q12

a) i)

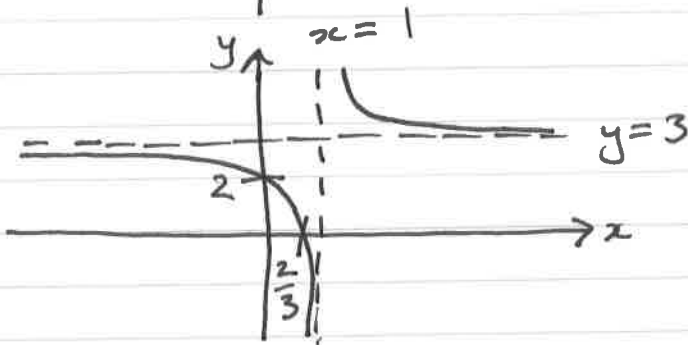


ii)



Shape ✓
intercepts ✓

iii)



Shape ✓
asymptotes ✓
intercepts ✓

b) i) $35^\circ + x = 180^\circ$ ✓ (Co-interior angles between // lines)
 $x = 145^\circ$ ✓
 $360^\circ - 320^\circ = 40^\circ$
 $145^\circ - 40^\circ = 105^\circ$ (Adjacent Angles) ✓

ii) $a^2 = b^2 + c^2 - 2bc \cos A$
 $a^2 = 3500^2 + 2000^2 - 2 \times 3500 \times 2000 \times \cos 105^\circ$ ✓
 $a = 4458 \text{ m}$ (to nearest m) ✓

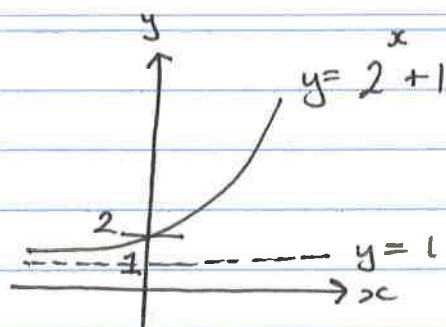
iii) $\frac{\sin A}{3500} = \frac{\sin 105^\circ}{4458}$
 $A = 49.3199\dots$ ✓

$\therefore$ Bearing

$= 360 - (49.31996 - 35)$
 $= 346 \text{ degrees to nearest degree.}$

Q13)

a) i)



asymptote, labelled; ✓
Shape + intercept; ✓

ii)

$$y > 1$$

✓

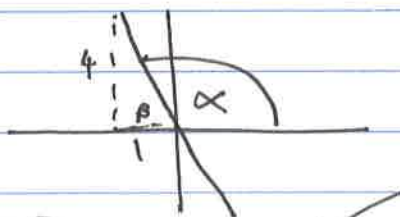
b)

$$m = -4$$

$$\tan \beta = 4$$

$$\beta = 75.96375 \dots$$

$$\angle = 104.036$$



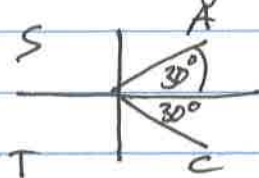
$$= 104^\circ \checkmark \text{ (to nearest degree) .}$$

c) $\cos 2\alpha = \frac{\sqrt{3}}{2}$ for $0^\circ \leq \alpha \leq 360^\circ$

let $u = 2\alpha$

$$\cos u = \frac{\sqrt{3}}{2} \quad \text{for} \quad 0^\circ \leq u \leq 720^\circ$$

related angle: $u = 30^\circ$ ✓



$$u = 30^\circ, 330^\circ, 390^\circ, 690^\circ$$

$$\alpha = 15^\circ, 115^\circ, 195^\circ, 345^\circ$$

(d) (i) $A(3, 9)$ $B(1, 1)$.

$$M = \left(\frac{3+1}{2}, \frac{9+1}{2} \right) \quad \checkmark$$

$M = (2, 5)$ centre of circle.

$$MB = \sqrt{(2-1)^2 + (5-1)^2} \quad \checkmark$$
$$= \sqrt{17}$$

$\therefore$ Radius is $\sqrt{17}$

ii) $y = x + 6$

$$(x-2)^2 + ((x+6)-5)^2 = 17 \quad \checkmark$$
$$x^2 - 4x + 4 + x^2 + 2x + 1 = 17$$

$$2x^2 - 2x + 5 = 17$$

$$2x^2 - 2x - 12 = 0$$

$$x^2 - x - 6 = 0$$

$$(x+2)(x-3) = 0$$

$$x = -2, \quad y = 4$$

or $x = 3$

$\therefore (-2, 4)$ is the other point of intersection.

Q14)

a) i) $A^3 + B^3 = (A+B)(A^2 - AB + B^2)$ ✓

ii) LHS = $\frac{(\sin x + \cos x)(\sin^2 x - \sin x \cos x + \cos^2 x)}{\sin x + \cos x}$ ✓

$$= \sin^2 x + \cos^2 x - \sin x \cos x$$

$$= 1 - \sin x \cos x$$
 ✓

$$= \text{RHS as required}$$

b) i) $f(x) = |x+3| - 2$

$$f(0) = |0+3| - 2$$

$$= 1$$

$$g(x) = -(x+2)^2 + 5$$

$$g(0) = -(0+2)^2 + 5$$

$$g(0) = 1$$
 ✓

(both)

ii) when $x < -3$ $f(x) = -(x+3) - 2$

$$-(x+3) - 2 = -(x+2)^2 + 5$$
 ✓

$$-x - 3 - 5 = -(x^2 + 4x + 4) + 5$$

$$-x - 5 = -x^2 - 4x + 1$$

$$x^2 + 3x - 6 = 0$$

$$x = \frac{-3 \pm \sqrt{9 - 4 \times 1 \times -6}}{2 \times 1}$$

$$x = \frac{-3 \pm \sqrt{33}}{2}$$

as $x < -3$, $x = \frac{-3 - \sqrt{33}}{2}$ ✓

$$f\left(\frac{-3 - \sqrt{33}}{2}\right) = \left|\frac{-3 - \sqrt{33}}{2} + 3\right| - 2$$

$$= \left|\frac{3 - \sqrt{33}}{2}\right| - 2$$

$$= \frac{\sqrt{33} - 7}{2}$$
 ✓

$$\left(\frac{-3 - \sqrt{33}}{2}, \frac{\sqrt{33} - 7}{2}\right)$$

iii) $\frac{-3 - \sqrt{33}}{2} \leq x \leq 0$ ✓

$$c) i) M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right) = (6, 7).$$

$$A(h, k).$$

let B have coordinates $B(x_1, y_1)$.

$$\left. \begin{array}{l} 6 = \frac{x_1 + h}{2} \\ 12 = x_1 + h \\ x_1 = 12 - h \\ \\ 7 = \frac{y_1 + k}{2} \\ 14 = y_1 + k \\ y_1 = 14 - k. \end{array} \right\} \checkmark \text{ (or equivalent)}$$

$$\therefore B(12 - h, 14 - k)$$

$$ii) A(h, k) \text{ lies on line } 2x + 5y - 15 = 0.$$

$$\therefore \textcircled{1} 2h + 5k - 15 = 0$$

$$B(12 - h, 14 - k) \text{ lies on line } 3x - 2y + 5 = 0$$

$$\therefore \textcircled{2} 3(12 - h) - 2(14 - k) + 5 = 0$$

$$36 - 3h - 28 + 2k + 5 = 0$$

$$-3h + 2k + 13 = 0.$$

$$\textcircled{1} \times 3 + \textcircled{2} \times 2:$$

$$6h + 15k - 45 = 0$$

$$-6h + 4k + 26 = 0$$

$$19k - 19 = 0$$

$$k = 1$$

$$h = 5$$

$$\} \checkmark$$

sub into $\textcircled{1}$

$$\therefore A(5, 1) \quad M(6, 7)$$

$$y - 1 = \frac{7 - 1}{6 - 5}(x - 5)$$

$$y = 6x - 29 \quad \checkmark$$